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A
PROJECT REPORT ON

Transistor-MOSFET Based IR Object Detection and Dual Alert System

SDG 9: Industry, Innovation and Infrastructure

SUBMITTED TO AISSMS'S INSTITUTE OF INFORMATION TECHNOLOGY, PUNE
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND TELECOMMUNICATION BY

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ACADEMIC YEAR: 2025-26



CERTIFICATE

This is to certify that Project Report entitled

“Transistor-MOSFET Based IR Object Detection and Dual Alert System”

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is the record of bonafide work carried out by them in partial fulfillment of the requirement for the award of the Degree of **Bachelor of Technology (Electronics and Telecommunication)**, as prescribed by the AISSMS Institute of Information Technology in the Academic Year 2025-2026.

This project report has not been earlier submitted to any other Institute or University for the award of any degree.

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Monesh Borole

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ABSTRACT

Object detection systems are widely used in applications such as security, automation, and safety systems, where non-contact sensing and quick response are essential. Infrared (IR) sensors are commonly preferred for such applications due to their low cost, simplicity, and fast operation. However, many existing object detection systems rely on microcontrollers and software-based processing, which increases system complexity, cost, and power consumption. Additionally, such systems may introduce delays due to processing time and often require programming knowledge, making them less suitable for simple and low-cost implementations.

In this project, a hardware-based object detection and alert system is designed and implemented without the use of a microcontroller. The system utilizes an IR sensor operating on active-low logic to detect the presence of an object. The sensor output is processed using a BC547 transistor for signal inversion, ensuring proper logic compatibility with the switching stage. A pull-up resistor is used to maintain a stable voltage at the MOSFET gate, preventing floating conditions. An IRFZ44N MOSFET is employed as a low-side switch to control the output load, which consists of an LED and a buzzer for visual and audio indication.

The developed system successfully detects objects and activates the alert mechanism with minimal delay and reliable performance. The results demonstrate that the proposed design achieves efficient object detection with low power consumption, fast response time, and a simple, cost-effective hardware configuration, making it suitable for various real-world applications.



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CHAPTER 1

INTRODUCTION

1.1 Background of the Project

Object detection systems are widely used in modern electronics for applications such as security, automation, and safety. With increasing demand for non-contact sensing and real-time response, infrared (IR) sensor-based systems have gained popularity due to their simplicity, low cost, and fast operation. These systems detect the presence of objects without physical contact and are commonly used in industrial, domestic, and robotic applications. Efficient switching mechanisms are also required to control output devices reliably, making the integration of sensors with electronic switching components an important area of study.

1.2 Problem Statement

Although various object detection systems are available, many of them rely on microcontrollers and software-based processing, which increases system complexity, cost, and power consumption. Additionally, improper interfacing between sensors and output devices can lead to unstable operation, signal mismatch, and inefficient switching. The problem addressed in this project is to design a simple, reliable, and cost-effective object detection system that eliminates the need for a microcontroller while ensuring stable operation and efficient control of output devices.

1.3 Motivation / Need for the Project

The motivation for this project arises from the need for simple and efficient hardware-based solutions in sensing and automation applications. Many real-world systems require fast response and low power consumption without involving complex programming. This project is undertaken to develop a practical system that demonstrates how basic electronic components such as sensors, transistors, and MOSFETs can be effectively combined to achieve reliable object detection and alert functionality.

1.4 Objectives of the Project

The main objectives of this project are:

- To study the working principle of IR-based object detection systems
- To design a hardware-based detection system without using a microcontroller
- To implement signal inversion using a transistor
- To develop a MOSFET-based switching circuit for controlling output devices.



1.5 Scope of the Project

The scope of this project is limited to the design and implementation of an IR-based object detection and alert system using discrete electronic components. The system focuses on detecting the presence of an object and activating an LED and buzzer as output indicators. Advanced features such as long-range detection, wireless communication, or microcontroller-based processing are not included. The project considers constraints such as cost, simplicity, and ease of implementation.

1.6 Methodology Overview

The project follows a structured approach starting with understanding the problem and selecting appropriate components. The system is then designed by dividing it into functional stages, including detection, signal processing, and switching. The circuit is implemented using an IR sensor, transistor, MOSFET, and supporting components. Finally, the system is tested under different conditions to verify its performance and reliability.

1.7 Organization of the Report

This report is organized into multiple sections. Chapter 1 presents the introduction, background, and objectives of the project. Chapter 2 covers the review of existing systems and their limitations. Chapter 3 explains the system design, methodology, and hardware implementation. Chapter 4 presents the results and analysis of the system performance. Chapter 5 concludes the project and suggests possible future improvements.



CHAPTER 2

LITERATURE REVIEW

[1] Motion Detection System Using Passive Infrared Technology

This paper explains the working of PIR (Passive Infrared) sensors used for motion detection systems. It highlights that PIR sensors detect changes in infrared radiation caused by human or object movement and are widely used in security and automation applications. The study demonstrates real-time detection capability with low power consumption and simple circuitry. The conclusion states that PIR-based systems are highly effective for intrusion detection and alert systems.

[2] Object Recognition with a Smart Low-Cost Active Infrared Sensor Array (2005)

This paper presents an active IR sensor array system for object recognition. It shows that multiple IR sensors arranged in an array improve detection accuracy and spatial resolution compared to a single sensor. The system is cost-effective and suitable for robotics and short-range detection tasks. The study concludes that IR arrays provide better reliability in structured detection environments.

[3] Learning of Embedded System Design, Simulation and Implementation

This paper focuses on embedded system design methodology, including simulation and hardware implementation techniques. It explains how embedded systems integrate sensors, processing units, and output devices for real-world applications. The study emphasizes systematic design flow from requirement analysis to implementation. It concludes that embedded systems provide efficient solutions for automation and control applications.

[4] A Review Paper on Infrared Sensor

This paper provides an overview of infrared sensor technology, including working principles, types, and applications. It explains that IR sensors are widely used in automation, robotics, and security systems due to their low cost and ease of use. The study concludes that IR sensors are effective for short-range, non-contact detection systems.



Summary of All Papers

The reviewed literature collectively highlights the importance of infrared sensing and embedded system design in modern automation applications. Papers [1], [2], and [4] establish that IR-based systems are widely used for short-range detection due to their simplicity, low cost, and fast response time. Paper [1] specifically focuses on motion detection using PIR sensors, while Paper [2] improves detection accuracy using IR sensor arrays. Paper [4] provides a foundational understanding of IR sensor technology and its applications.

Paper [3] complements these studies by explaining embedded system design principles, showing how sensors and electronic components can be integrated into complete functional systems. However, it is more theoretical and does not focus deeply on IR hardware implementation.

Overall, the literature confirms that IR-based sensing combined with simple embedded hardware design is highly suitable for low-cost, real-time object detection systems. This directly supports the design approach of the proposed project, which uses IR sensing, transistor signal processing, and MOSFET-based switching without a microcontroller.



CHAPTER 3

METHODOLOGY

- **Component Selection**

The required components such as IR sensor, BC547 transistor, IRFZ44N MOSFET, resistors, LED, and buzzer were selected based on availability, compatibility, and circuit requirements.

- **Circuit Design**

A circuit was designed to interface the IR sensor with the output devices. The design included three main stages: detection (IR sensor), signal conditioning (transistor), and switching (MOSFET). Proper resistor values were chosen to ensure safe and stable operation.

- **IR Sensor Integration**

The IR sensor was connected to a 5V power supply. Its output pin was used as the input signal for the transistor stage to detect object presence based on reflected infrared radiation.

- **Signal Inversion using Transistor**

A BC547 NPN transistor was used to invert the active-low output of the IR sensor. A $1k\Omega$ resistor was connected to the base to limit current and protect the transistor.

- **Pull-up Configuration**

A $10k\Omega$ pull-up resistor was connected between the MOSFET gate and the supply voltage to maintain a defined HIGH state when the transistor is OFF, preventing floating conditions.

- **MOSFET Switching Implementation**

An IRFZ44N MOSFET was configured as a low-side switch. The gate received the processed signal, while the drain was connected to the load and the source to ground.

- **Output Load Connection**

The LED (with current-limiting resistor) and buzzer were connected to the MOSFET drain, allowing them to be controlled simultaneously based on object detection.

- **Testing and Verification**

The system was tested by placing objects in front of the IR sensor. The output behavior (LED and buzzer activation) was observed and verified under different conditions.

- **Result Analysis**

The circuit successfully detected object presence and activated the alert system, confirming correct working of sensing, inversion, and switching stages.



CHAPTER 4

HARDWARE/SOFTWARE DESIGN

5.1 Hardware Design

5.1.a Block Diagram

The block diagram of the IR-based object detection and alarm system is shown below:

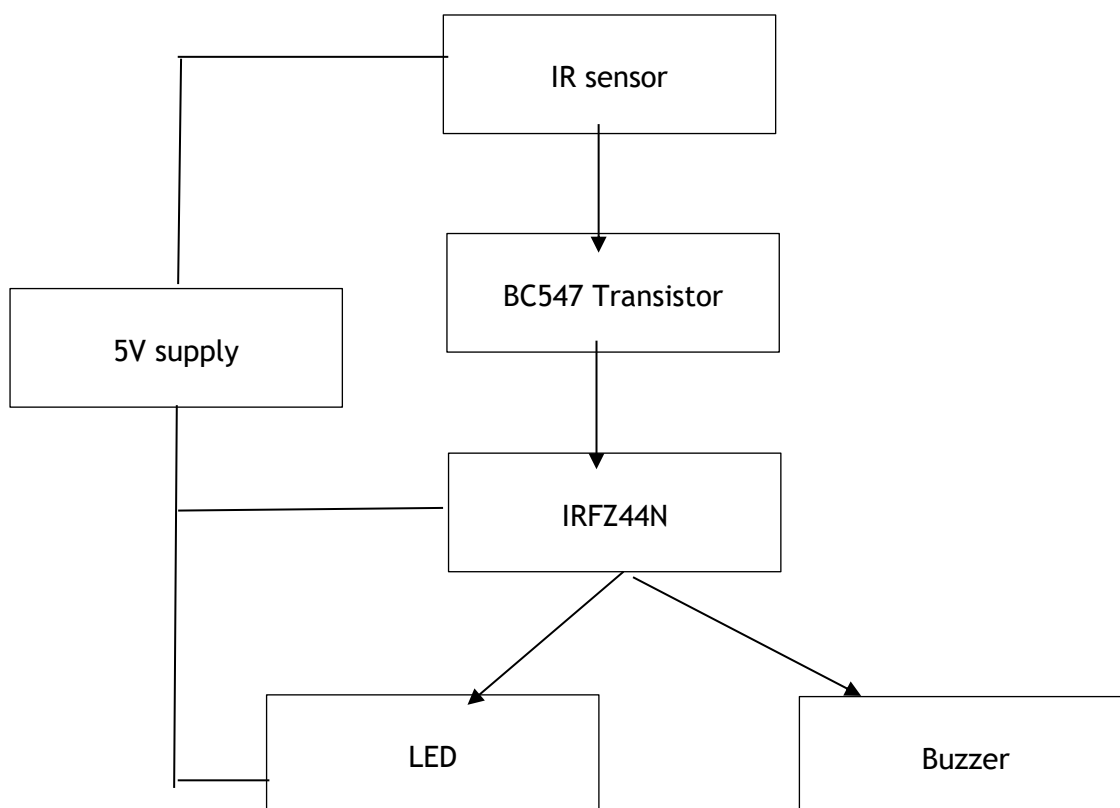


Fig 1 : Block diagram



5.1.b Technical specifications

Table 1: Technical specifications

Component	Specification	Function in Circuit
IR Sensor	Operating Voltage: 3.3V–5V DC Output: Digital (Active LOW) Type: Infrared proximity sensor	Detects object using infrared reflection and generates logic signal
BC547 Transistor	Type: NPN BJT Configuration: Common emitter	Used for signal inversion and switching control
Pull-up Resistor	Value: 10 k Ω	Provides stable HIGH logic level at MOSFET gate and prevents floating
IRFZ44N MOSFET	Type: N-channel enhancement MOSFET Gate Threshold: 2V–4V	Acts as low-side electronic switch to control load
LED	Forward Voltage: ~2V Current: 10–20 mA (with resistor)	Provides visual indication of object detection
Buzzer	Operating Voltage: 5V Type: Active buzzer	Provides audio alert when system is triggered

5.1.c Explanation of Block Diagram

- **IR Sensor** continuously emits infrared light and when an object is detected, the output goes LOW (active LOW behavior).
- **BC547 Transistor** inverts this signal when IR is LOW, transistor turns OFF and collector becomes open, and when IR is HIGH, transistor turns ON and pulls collector to GND.
- **Pull-up Resistor (10k Ω)** connects MOSFET gate to +5V and ensures the gate gets a clean HIGH or LOW signal without floating.
- **IRFZ44N MOSFET** acts as a low-side switch where gate HIGH turns it ON allowing current to flow, and gate LOW turns it OFF stopping current completely.
- **LED and Buzzer** are both connected to the MOSFET drain and activate together when MOSFET turns ON, giving visual and audio alert on object detection.



5.2 SOFTWARE DESIGN

5.2 a Overview

The simulation of this project was implemented using Proteus Professional 8. All components including FC-51 IR sensor, BC547 transistor, IRFZ44N MOSFET, resistors, LED, and buzzer were placed and configured as per their datasheet values. The IR sensor output was toggled to simulate real-world object detection and both conditions were verified to confirm correct circuit operation before hardware implementation.

5.b Circuit Simulation

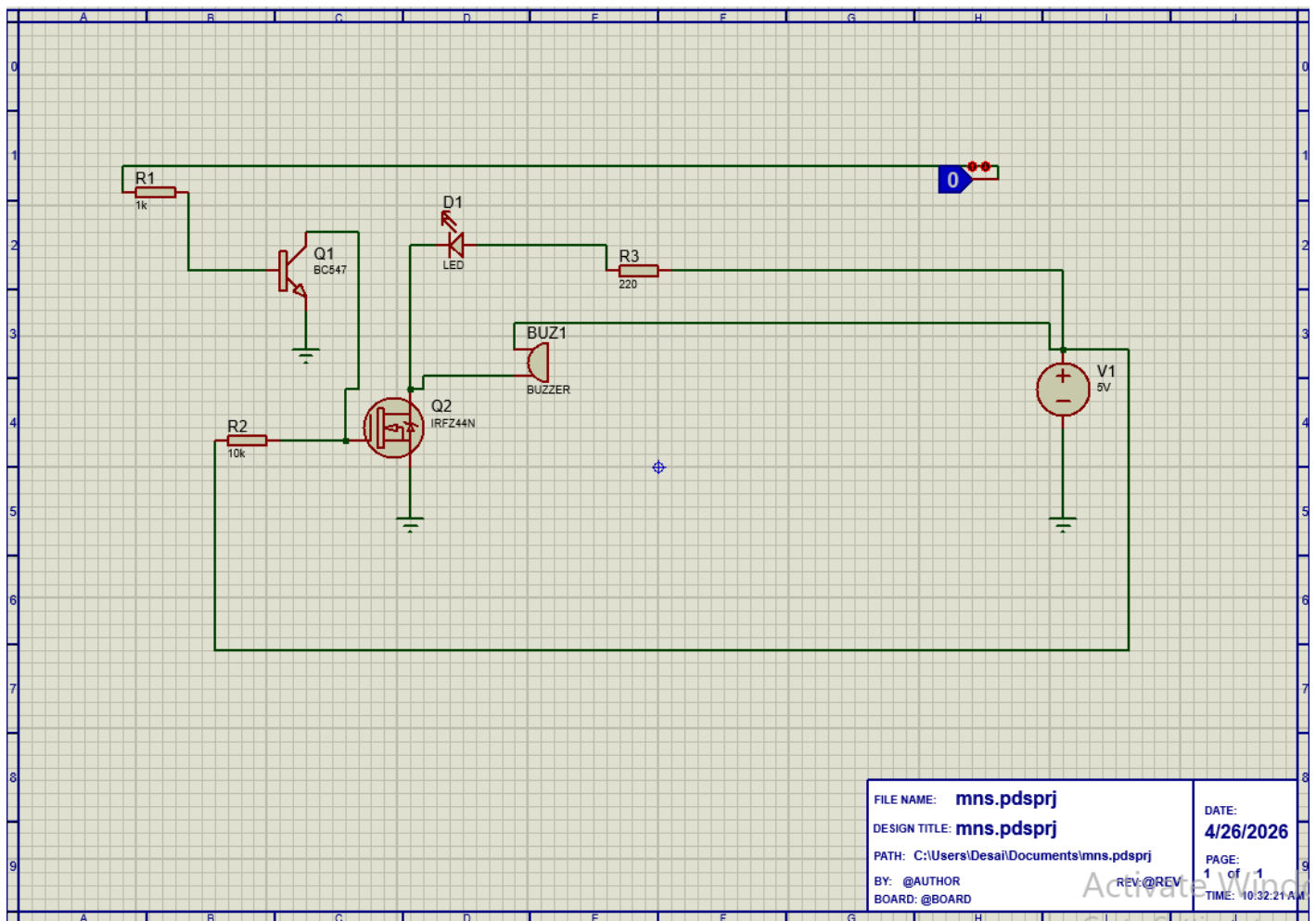


Fig 2: Proteus simulation



5.2.c PHASES OF SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

- Although this project is primarily hardware-based, the SDLC framework is applied to the overall project development process, which includes circuit design, simulation, implementation, and documentation:
- Planning Identifying the objective of the simulation — to verify the working of IR sensor based object detection circuit using BC547, IRFZ44N, LED and buzzer on Proteus Professional 8.
- Requirement Analysis Listing all components required in the simulation — FC-51 IR sensor, BC547 transistor, IRFZ44N MOSFET, 1k Ω , 10k Ω , 220 Ω resistors, LED, active buzzer and 5V power supply.
- System Design Designing the circuit schematic on Proteus by placing all components, making connections as per the circuit diagram and setting correct component values and properties.
- Implementation Running the simulation on Proteus, testing the circuit by toggling the IR sensor output and observing whether LED and buzzer respond correctly to object detection and no-object conditions.
- Testing Verifying all conditions, object detected (LED ON, Buzzer ON) and no object (LED OFF, Buzzer OFF) and checking that transistor inversion and MOSFET switching are working correctly in simulation.
- Maintenance Making corrections if any component behaves unexpectedly in simulation, adjusting resistor values if needed and re-verifying the circuit until output matches expected behavior.

CHAPTER 5

EXPERIMENTAL RESULTS AND ANALYSIS

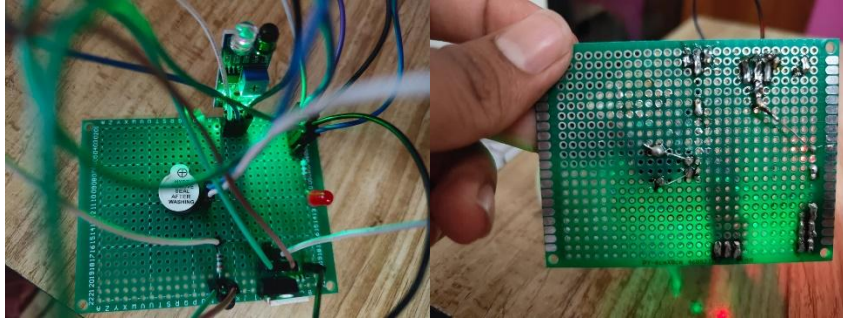


Fig 3: Hardware setup on zero PCB

Table 2: output analysis

Test Condition	IR output	Transistor State	MOSFET	LED	Buzzer	Final outcome
No object present	HIGH	ON	OFF	OFF	OFF	System idle
Object presented	LOW	OFF	ON	ON	ON	Alert activated
Object removed	HIGH	ON	OFF	OFF	OFF	System setback to idle



CHAPTER 6

CONCLUSIONS AND FUTURE SCOPE

The IR-based object detection and alert system successfully demonstrate a simple, reliable, and cost-effective solution for short-range non-contact object sensing. The system achieves its primary objectives: detecting the presence of an object using IR reflection and activating both a visual LED indicator and an auditory buzzer alert without the use of a microcontroller.

Key achievements of this project include:

- Successful implementation of a hardware-only IR detection circuit using BC547 transistor and IRFZ44N MOSFET switching
- Reliable object detection within the effective range of 2cm to 30cm using the FC-51 IR sensor
- Fast response time due to purely analog hardware implementation with no processing delay
- Low-cost design using minimal and easily available components like BC547, IRFZ44N, and FC-51
- Practical demonstration of low-side MOSFET switching relevant to power electronics applications

The project reinforces understanding of core electronics concepts including active LOW sensor behavior, signal inversion using NPN transistor, pull-up resistor for stable gate control, and MOSFET operation as an electronic switch for driving loads. The design is aligned with SDG Goal 9 by enabling affordable and innovative technology for security and automation applications.

FUTURE SCOPE:

- Integration of a microcontroller to enable advanced features such as object counting, data logging, and intelligent decision-making.
- Enhancement of detection accuracy and range by using multiple sensors or combining IR with other technologies like ultrasonic or vision-based systems.
- Addition of wireless communication modules (Bluetooth/Wi-Fi) for remote monitoring and alert notification in IoT-based applications.
- Implementation of power optimization techniques such as battery operation and low-power modes for portable and energy-efficient systems.
- Upgradation of the switching stage to handle higher power loads and integration into industrial automation and smart security systems.



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- [2] Object recognition with a smart low-cost active infrared sensor array, November 2005, Conference: Int. Conf. on Sensing Technology
- [3] Learning of Embedded System Design, Simulation and Implementation: A Technical Approach
Lukman Adewale Ajao, James Agajo, Jonathan Gana Kolo, Mutiu Adesina Adegboye, Yakubu Yusuf
- [4] A Review Paper on Infrared sensor Pooja Ajmera
- [5] [IRFZ44N datasheet](#)



APPENDIX

I. Total Cost of project

Component Name	Cost
MOSFET	60/-
Transistor	35/-
LED	5/-
Buzzer	15/-
Zero PCB	25/-
Miscellaneous	30/-
Total	170/-



II. Project Title Mapping with POs and PSOs

Project Title:

Gr. No	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
6														

Justification: